

ILE WALLIS, Wallis and Futuna

WMO: 917530

Lat: 13.238S

Lon: 176.199W

Elev: 79

StdP: 14.65

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	73.3	74.4	65.2	93.5	77.8	67.3	100.6	78.5	18.7	80.5	17.8	81.0	3.9	160	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	8.9	88.7	80.5	88.1	80.2	87.7	80.0	81.8	87.0	81.3	86.5	80.9	86.1	10.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	157.8	85.3	79.8	155.1	84.9	79.2	152.1	84.3	45.7	87.2	45.1	86.8	44.6	86.5	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.8	16.1	14.5	DB	70.5	89.5	2.6	0.8	68.7	90.0	67.1	90.5	65.7	91.0	63.8	91.6
			WB	66.3	82.6	2.8	0.8	64.3	83.2	62.7	83.6	61.1	84.1	59.1	84.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	82.0	82.0	82.2	82.2	81.6	81.3	80.5	80.6	81.1	81.2	81.4	82.0
	DBStd	1.73	1.76	1.76	1.56	1.65	1.65	1.57	1.64	1.62	1.58	1.62	1.71	1.57
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11500	993	896	997	965	980	940	947	948	933	967	943	991
	CDD65	6025	528	476	532	515	515	490	482	483	483	502	493	526
	CDH74	63414	5821	5232	5889	5711	5432	5056	4659	4651	4830	5124	5234	5775
(13)	CDH80	17398	1825	1645	1882	1834	1530	1205	935	937	1097	1262	1477	1769
(14)	Wind	WSAvg	7.9	7.2	7.4	6.7	6.9	7.4	9.1	9.3	9.5	8.3	7.2	7.0
(15) Precipitation	PrecAvg	124.0	14.5	11.3	12.0	11.7	8.7	6.4	8.1	6.5	7.8	11.7	11.8	13.4
	PrecMax	166.1	30.0	30.2	28.5	24.1	14.8	12.1	22.3	20.0	16.9	19.2	28.1	25.4
	PrecMin	91.9	4.5	1.0	1.7	3.7	2.2	1.4	2.4	0.4	0.5	3.1	3.7	5.3
	PrecStd	18.6	6.1	6.3	6.6	5.5	3.7	3.1	5.5	4.1	4.0	4.2	4.8	4.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.2	89.1	89.1	89.1	88.2	87.3	86.4	86.4	87.1	87.6	88.2	88.7
		MCWB	81.0	80.7	80.6	80.9	80.0	79.2	78.9	78.5	78.8	79.2	79.8	80.6
	2%	DB	88.2	88.2	88.2	88.2	87.6	86.4	85.9	85.9	86.3	86.8	87.5	87.9
		MCWB	80.4	80.3	80.2	80.4	79.8	78.9	78.6	78.2	78.4	78.9	79.5	80.2
	5%	DB	87.4	87.6	87.6	87.6	86.8	85.9	85.2	85.2	85.6	86.1	86.8	87.2
		MCWB	80.0	80.0	79.9	80.2	79.4	78.8	78.2	77.8	78.1	78.7	79.2	79.9
	10%	DB	86.6	86.7	86.8	86.8	85.9	85.0	84.4	84.5	85.0	85.4	86.0	86.4
		MCWB	79.7	79.7	79.7	79.8	78.9	78.3	77.7	77.4	77.8	78.3	78.9	79.5
	0.4%	WB	82.6	82.3	82.0	81.9	81.6	80.9	80.4	80.1	80.6	80.9	81.3	81.9
		MCDB	87.7	87.5	87.1	87.3	86.5	85.8	85.2	85.0	85.3	85.7	86.5	87.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	81.6	81.4	81.3	81.3	80.8	80.1	79.6	79.2	79.8	80.0	80.5	81.1
		MCDB	86.7	86.7	86.5	86.8	86.1	84.9	84.4	84.3	84.7	85.1	86.1	86.4
	5%	WB	80.8	80.8	80.7	80.7	80.1	79.5	79.0	78.7	79.1	79.5	79.9	80.6
		MCDB	85.9	86.1	86.0	86.4	85.2	84.3	83.8	83.8	84.1	84.5	85.2	86.0
	10%	WB	80.2	80.1	80.1	80.1	79.5	78.9	78.3	78.0	78.5	78.9	79.3	80.0
		MCDB	85.0	85.2	85.2	85.4	84.4	83.6	82.9	82.9	83.4	84.0	84.4	85.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	8.7	8.9	8.9	8.2	7.0	7.0	7.3	7.5	8.0	8.5	8.5
		MCDBR	9.5	9.6	9.7	9.6	9.3	7.8	8.0	8.0	8.1	8.8	9.5	9.5
		MCWBR	4.2	4.2	4.3	4.4	4.4	3.9	4.0	4.1	4.0	4.2	4.3	4.3
	5% WB	MCDBR	8.9	8.9	9.0	9.1	8.6	7.5	7.7	7.9	7.8	8.3	8.9	8.9
		MCWBR	4.3	4.3	4.4	4.4	4.4	4.0	4.1	4.2	4.0	4.4	4.4	4.4
(40) Clear-Sky Solar Irradiance	taub	0.385	0.377	0.376	0.384	0.387	0.398	0.400	0.399	0.405	0.388	0.382	0.383	
	taud	2.642	2.673	2.678	2.654	2.631	2.580	2.564	2.565	2.547	2.607	2.640	2.648	
	Ebn at Noon	304	304	298	286	275	266	268	277	285	297	303	304	
	Edh at Noon	32	30	30	29	28	29	30	31	33	32	32	31	
(44) All-Sky Solar Radiation	RadAvg	1830	1801	1798	1637	1456	1366	1447	1655	1840	1884	1864	1866	
(45)	RadStd	214	200	117	139	109	79	68	75	93	117	120	103	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.31	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page